IGCSE 0607 Practice Questions — E5.6

Circle theorems I

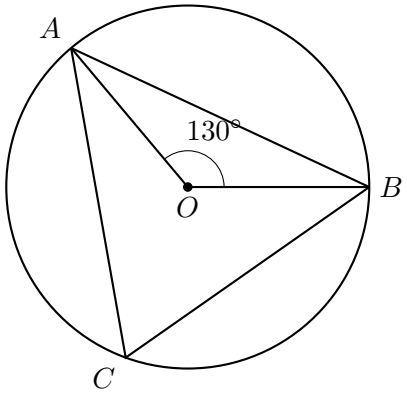
Total Questions: 45 — **Total Marks:** 253 Marks

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- **E5.6** Circle theorems I
  → **Paper 2 (Non-Calculator)** *(24 Qs)* · **Paper 4 (Calculator)** *(21 Qs)*

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1 Points A, B and C lie on a circle, centre O.
Angle $AOB = 130°$.



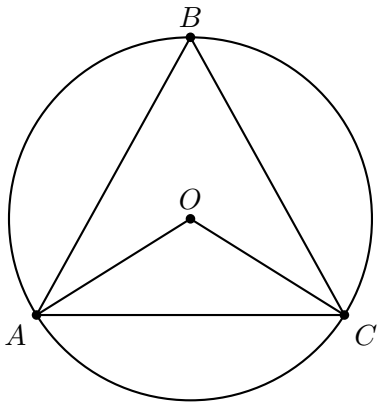
(a) Find the size of angle ACB.

..... ° **[1]**

(b) Find the size of angle OAB.

..... ° **[2]**

2 A, B and C are points on a circle, center O.
Angle $ABC = 58°$.



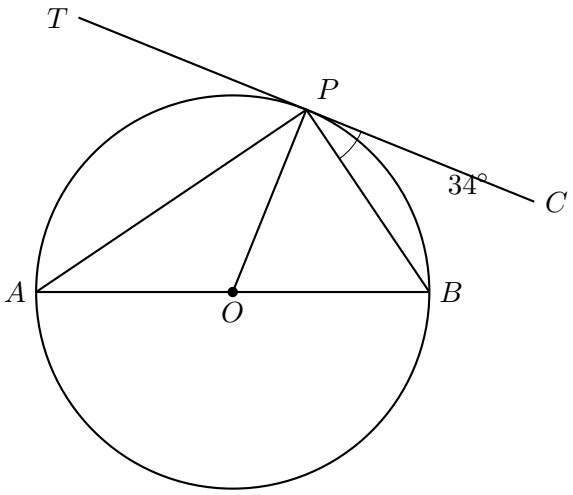
(a) Work out the obtuse angle AOC.

Angle AOC = ... ° **[1]**

(b) Work out angle OAC.

Angle OAC = ... ° **[2]**

3 The diagram shows a circle, centre O.
A, B and P are points on the circumference.
AB is a diameter of the circle.
The line TC is a tangent to the circle at the point P.
Angle $BPC = 34°$.



NOT TO SCALE

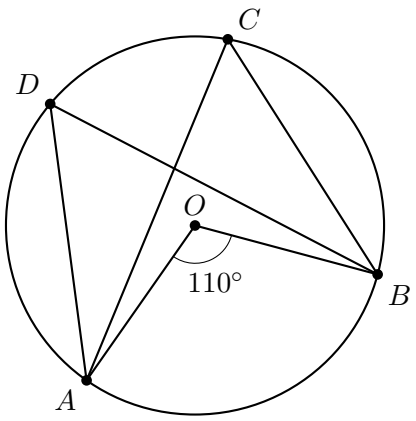
(a) Write down the value of angle PAB.

Angle $PAB =$ ..... ° **[1]**

(b) Work out the value of angle POB.

Angle $POB =$ ..... ° **[2]**

4 A, B, C and D are points on a circle, centre O.
Angle $AOB = 110°$.



NOT TO SCALE

(a) Find the size of angle ACB.

Angle $ACB =$ ..... ° **[1]**

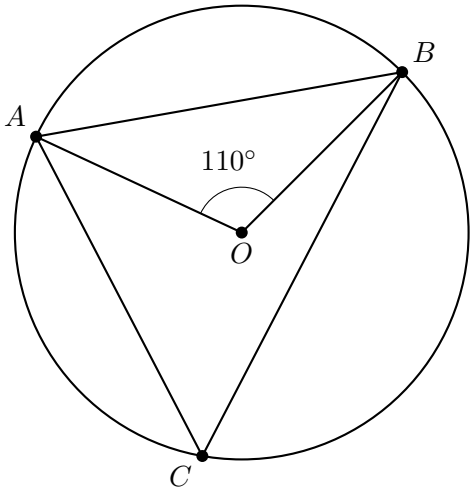
(b) Find the size of angle ADB.

Angle $ADB =$ ..... ° **[1]**

(c) State the geometrical reason for your answer to part (b).

..... **[1]**

5 The diagram shows a circle, centre O.
A, B and C are points on the circumference.
Angle $AOB = 110°$.



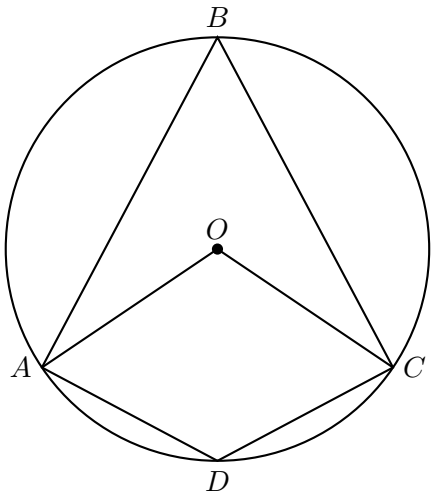
(a) **(i)** Calculate the size of angle ACB.

Angle ACB = ... ° **[1]**

(ii) Calculate the size of angle OAB.

Angle OAB = ... ° **[2]**

6 A, B, C and D are points on a circle, center O.
Angle $ABC = 56°$.



(a) Calculate the obtuse angle AOC.

..... ° **[1]**

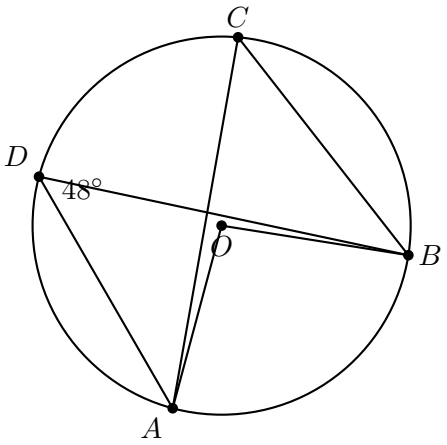
(b) Calculate angle ADC.

..... ° **[1]**

(c) State the geometric reason for your answer to part (b).

..... **[1]**

7 A, B, C and D are points on a circle, centre O.
Angle $ADB = 48°$.



(a) **(i)** Find angle ACB.

Angle $ACB =$ ..... ° **[1]**

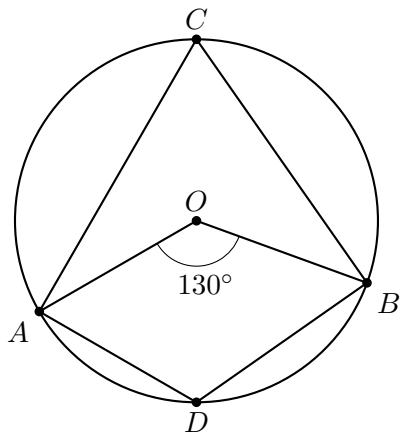
(ii) Give a geometrical reason for your answer to part (a)(i).

..... **[1]**

(b) Find angle AOB.

Angle $AOB =$ ..... ° **[1]**

8 Points A, B, C and D lie on the circumference of a circle, center O.
Angle $AOB = 130°$.



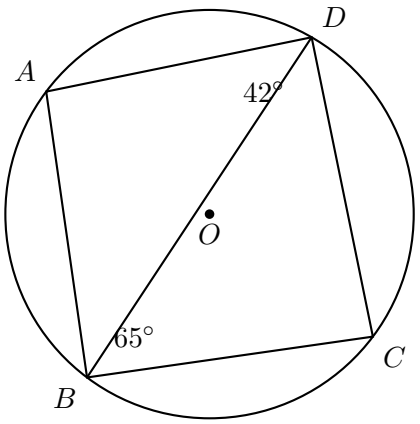
(a) Calculate angle ACB.

..... ° **[1]**

(b) Calculate angle ADB.

..... ° **[2]**

9 A, B, C and D are points on the circumference of a circle, centre O.
Angle $BDC = 42°$ and angle $CBD = 65°$.



(a) Find angle BAC.

Angle $BAC =$ ..... ° **[1]**

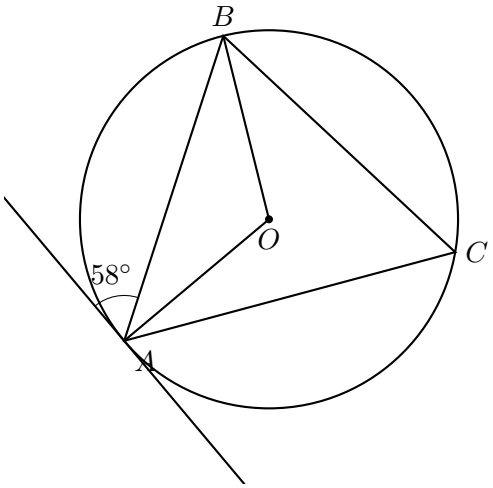
(b) Find angle BCD.

Angle $BCD =$ ..... ° **[1]**

(c) Find angle BAD.

Angle $BAD =$ ..... ° **[2]**

10 The points A, B, and C lie on a circle, centre O.
The line SAT is a tangent to the circle at A.
Angle $SAB = 58°$.



(a) Find angle ACB and state a geometrical reason for your answer.

(i) Angle ACB

..... ° **[1]**

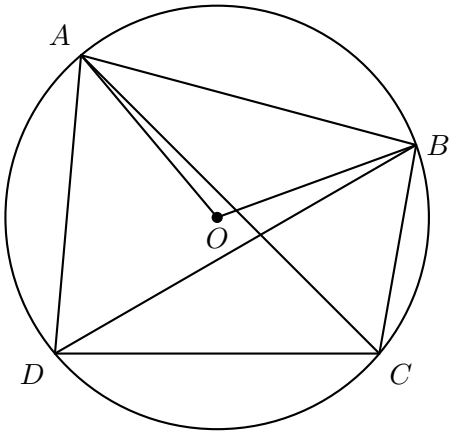
(ii) Reason

..... **[1]**

(b) Find angle OBA.

..... ° **[2]**

11 The diagram shows a circle with centre O.
The points A, B, C and D lie on the circumference of the circle.
Angle $AOB = 110°$ and angle $BDC = 30°$.



(a) Calculate angle ACB.

..... ° **[1]**

(b) State the geometrical reason for your answer to part (a).

..... **[1]**

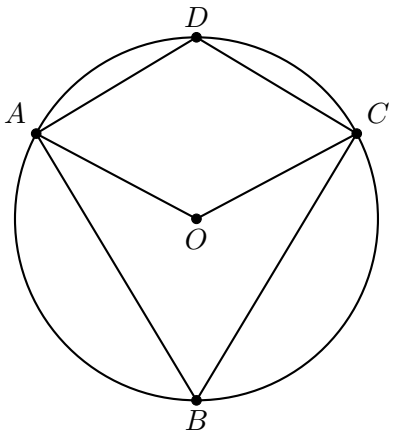
(c) Calculate angle BAC.

..... ° **[1]**

(d) Calculate angle OAB.

..... ° **[2]**

12 A, B, C and D are points on the circumference of a circle, centre O.
Angle $AOC = 124°$.



(a) Calculate angle ABC.

..... ° **[2]**

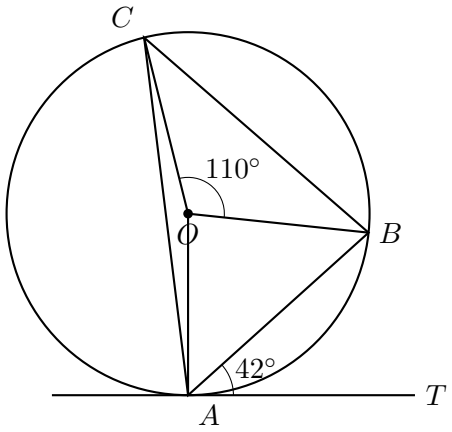
(b) Calculate angle ADC.

..... ° **[2]**

(c) Calculate angle OAC.

..... ° **[2]**

13 A, B and C are points on a circle, centre O.
TA is a tangent to the circle at A.
C is a point on the major arc AB.
Angle $TAB = 42°$ and angle $BOC = 110°$.



(a) Write down the value of

(i) angle BCA,

Angle $BCA =$ ..... ° **[1]**

(ii) angle BAC.

Angle $BAC =$ ..... ° **[1]**

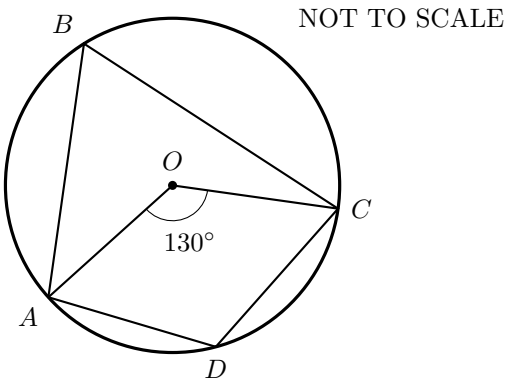
(b) Calculate angle OAB.

Angle $OAB =$ ..... ° **[2]**

(c) Calculate angle ABC.

Angle $ABC =$ ..... ° **[2]**

14 The diagram shows a circle, centre O, with points A, B, C, and D lying on the circumference. Angle $AOC = 130°$ and angle $OAB = 22°$.



(a) Find the size of angle ABC and give a geometrical reason for your answer.

Angle $ABC =$ ..... , Reason: ..... **[2]**

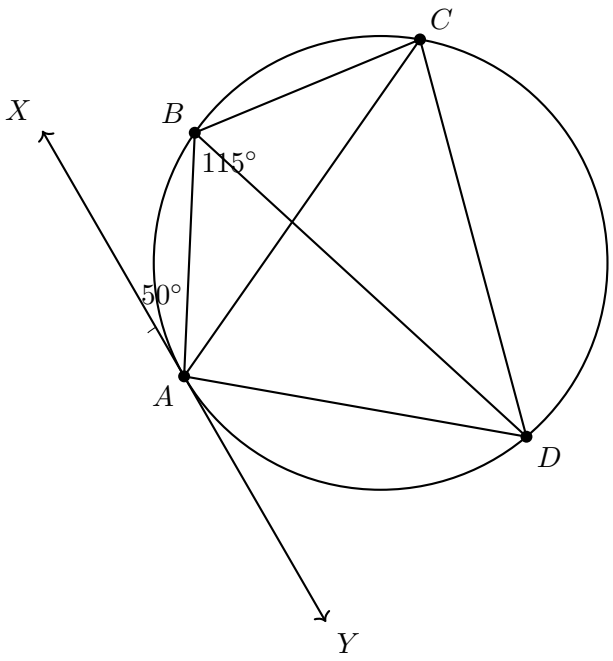
(b) Find the size of angle ADC and give a geometrical reason for your answer.

Angle $ADC =$ ..... , Reason: ..... **[2]**

(c) Calculate the size of angle OCB.

Angle $OCB =$ ..... ° **[2]**

15 The diagram shows a cyclic quadrilateral $ABCD$.
The line XY is a tangent to the circle at A.
Angle $XAD = 50°$ and angle $ABC = 115°$.



(a) **(i)** Write down the size of angle ACD.

Angle $ACD =$ ..... ° **[1]**

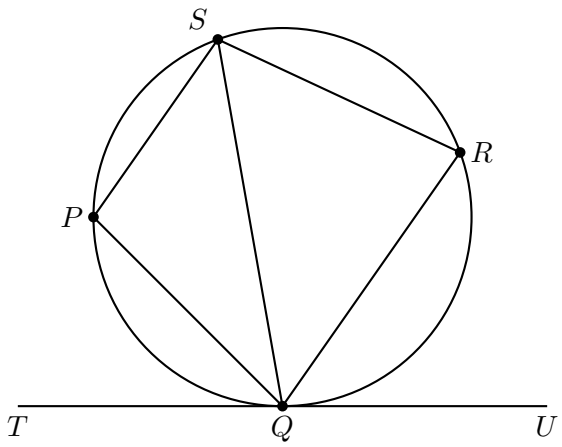
(ii) Calculate the size of angle ADC.

Angle $ADC =$ ..... ° **[1]**

(iii) Calculate the size of angle CAD.

Angle $CAD =$ ..... ° **[2]**

16 P, Q, R and S are points on a circle.
TQU is a tangent to the circle at Q.
Angle $RQU = 55°$ and Angle $QPS = 100°$.



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(a) Find the size of angle QSR.

Angle $QSR =$ ..... ° **[1]**

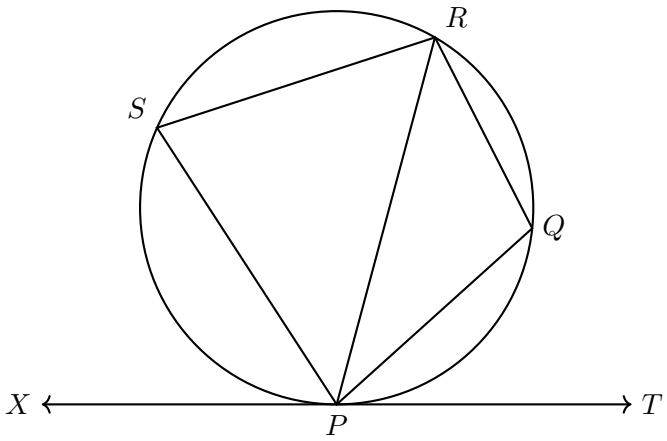
(b) Find the size of angle QRS.

Angle $QRS =$ ..... ° **[2]**

(c) Calculate the size of angle SQR.

Angle $SQR =$ ..... ° **[2]**

17 Points P, Q, R and S lie on a circle.
The straight line XPT is a tangent to the circle at P.
Angle $QPT = 42°$, angle $PQR = 105°$ and angle $SPR = 48°$.



(a) Find the size of angle PRQ.

..... ° **[1]**

(b) Find the size of angle PSR.

..... ° **[1]**

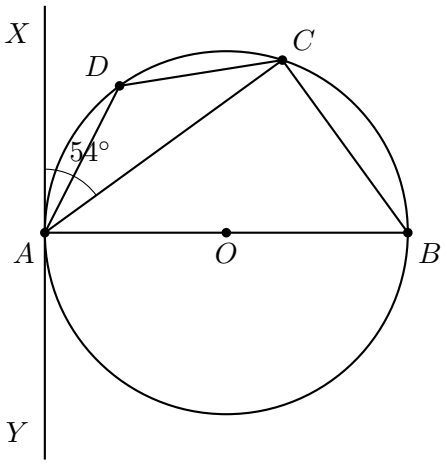
(c) Find the size of angle PRS.

..... ° **[2]**

(d) Find the size of angle XPS.

..... ° **[1]**

18 In the diagram, AB is a diameter of the circle, center O.
The line XAY is a tangent to the circle at A.
C and D are points on the circle.
Angle $XAC = 54°$.



(a) **(i)** Write down the size of angle ACB.

..... ° **[1]**

(ii) Give a geometric reason for your answer.

..... **[1]**

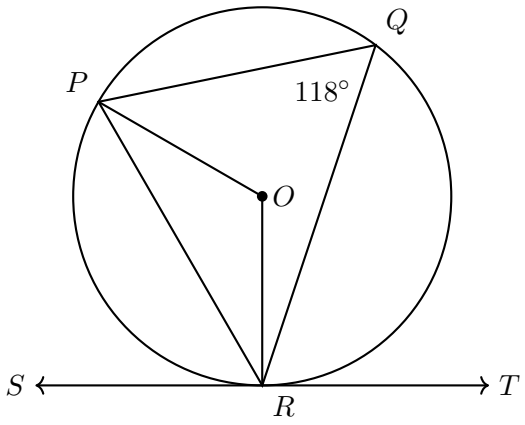
(b) Find angle ABC.

..... ° **[1]**

(c) Find angle ADC.

..... ° **[2]**

19 P, Q and R are points on a circle, centre O.
ST is a tangent to the circle at R.
Angle $PQR = 118°$.



(a) Find reflex angle POR.

Reflex angle $POR =$ ..... ° **[2]**

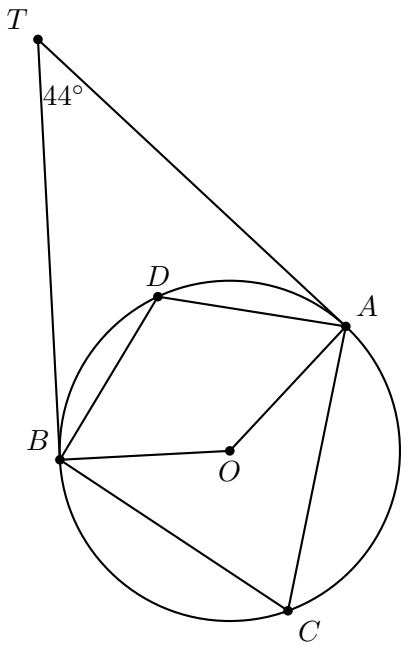
(b) Find obtuse angle POR.

Obtuse angle $POR =$ ..... ° **[1]**

(c) Find angle PRS.

Angle $PRS =$ ..... ° **[2]**

20 A, B, C and D are points on a circle, centre O.
TA and TB are tangents to the circle at A and B.
Angle $ATB = 44°$.



(a) Calculate angle AOB.

Angle $AOB =$ ..... ° **[2]**

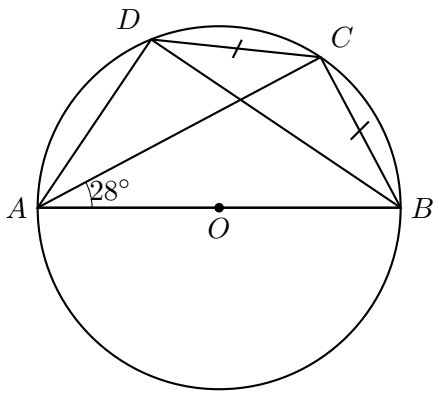
(b) Calculate angle ACB.

Angle $ACB =$ ..... ° **[1]**

(c) Calculate angle ADB.

Angle $ADB =$ ..... ° **[2]**

21 A, B, C and D are points on a circle.
AB is a diameter of the circle.
$CD = CB$ and angle $CAB = 28°$.



(a) (i) Find angle CBA.

Angle $CBA =$ ..... ° **[1]**

(ii) Give a geometrical reason for your answer.

..... **[1]**

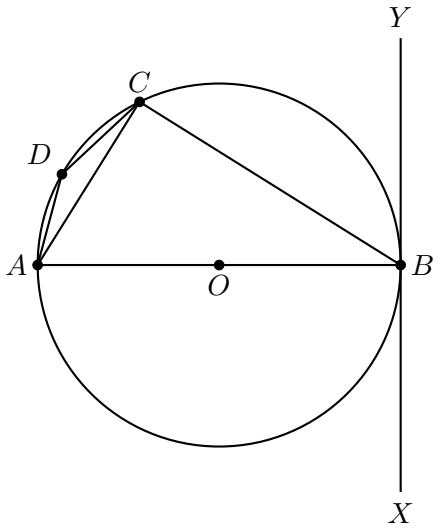
(b) Find angle CAD.

Angle $CAD =$ ..... ° **[1]**

(c) Find angle ADC.

Angle $ADC =$ ..... ° **[2]**

22 A, B, C and D lie on a circle, centre O.
AB is a diameter of the circle.
The line XY is a tangent to the circle at B.
Angle $CBY = 58°$.



(a) **(i)** Find angle BAC.

..... ° **[1]**

(ii) State the circle theorem that gives this angle directly from angle CBY.

..... **[1]**

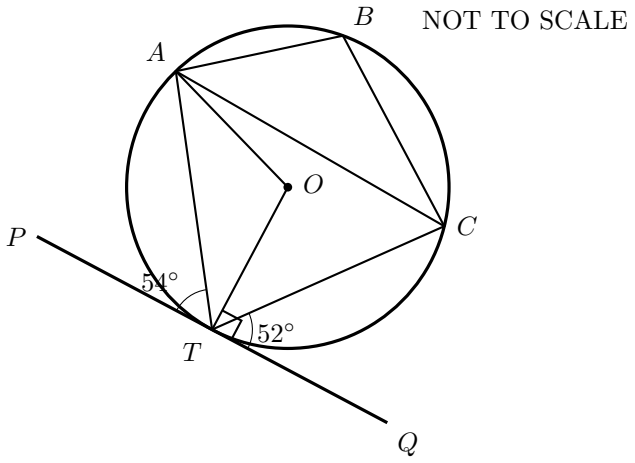
(b) Calculate angle ABC.

..... ° **[3]**

(c) Calculate angle ADC.

..... ° **[2]**

23 The diagram shows a circle, centre O, with points A, B, C, and T lying on the circumference.
The straight line PTQ is a tangent to the circle at T.
Angle $PTA = 54°$, angle $QTC = 52°$, and angle $BCT = 86°$.



(a) Find the size of angle TAC and give a geometrical reason for your answer.

Angle $TAC =$ ..... , Reason: ..... **[2]**

(b) Calculate the size of

(i) angle ABC,

Angle $ABC =$ ..... ° **[2]**

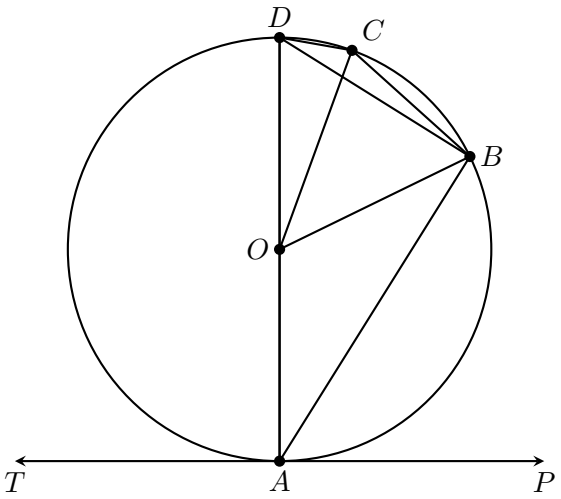
(ii) angle ACB,

Angle $ACB =$ ..... ° **[2]**

(iii) angle OAT.

Angle $OAT =$ ..... $^\circ$ **[3]**

24 A, B, C and D are points on a circle, center O.
AD is a diameter of the circle.
PAT is a tangent to the circle at A.
Angle $PAB = 58°$ and angle $BDC = 22°$.



(a) Calculate angle ADB.

Angle ADB = ... ° **[2]**

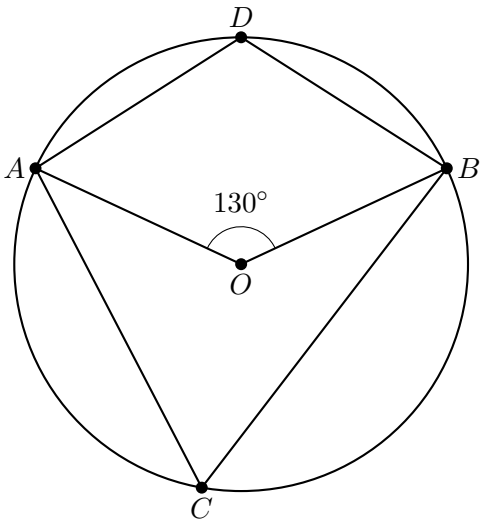
(b) Calculate angle ABC.

Angle ABC = ... ° **[2]**

(c) Calculate angle OBC.

Angle OBC = ... ° **[3]**

25 The points A, B, C, and D lie on a circle with centre O.
Angle $AOB = 130°$.



(a) Calculate angle ACB.

Angle $ACB =$ ..... ° **[1]**

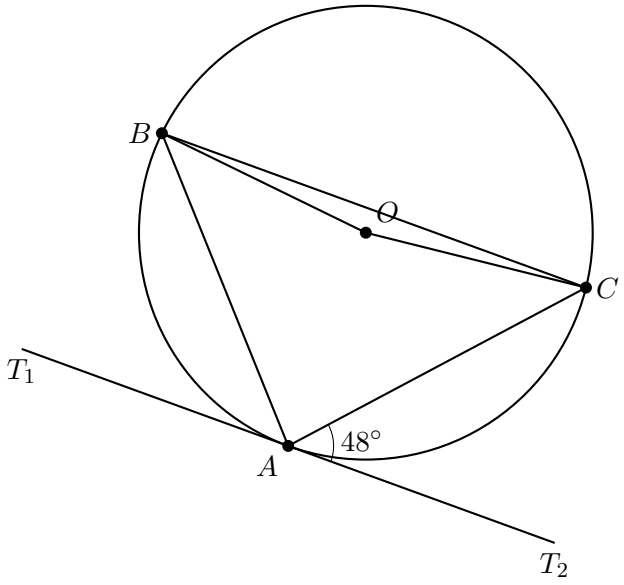
(b) Calculate angle ADB.

Angle $ADB =$ ..... ° **[1]**

(c) Give a geometrical reason for your answer to part (b).

..... **[1]**

26 The points A, B, and C lie on a circle with centre O.
The straight line T_1AT_2 is a tangent to the circle at A.
The chord BC is parallel to the tangent T_1AT_2.
Angle $CAT_2 = 48°$.



(a) Find angle ABC.

Angle ABC = ... ° **[1]**

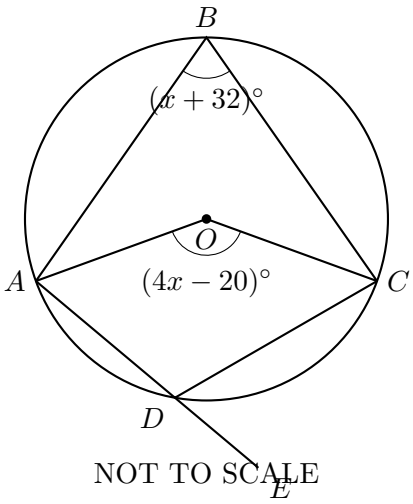
(b) Calculate angle BAC.

Angle BAC = ... ° **[2]**

(c) Calculate angle OBC.

Angle OBC = ... ° **[2]**

27 A, B, C and D lie on a circle, centre O.
Angle $ABC = (x + 32)°$ and the obtuse angle $AOC = (4x - 20)°$.
The line AD is extended to the point E.



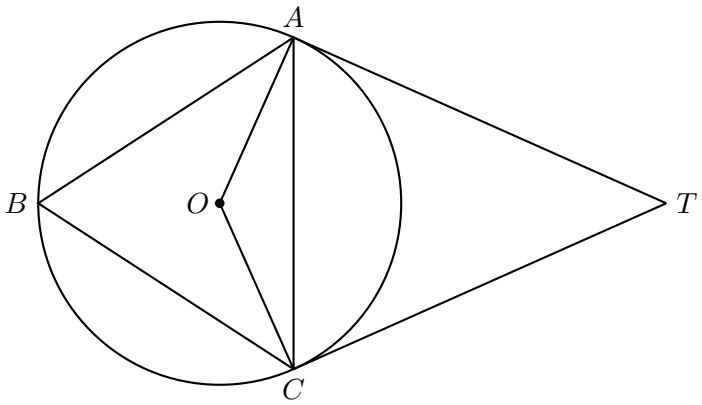
(a) Find the value of x.

$x =$ ..... **[3]**

(b) Work out the size of angle CDE.

Angle $CDE =$ ..... ° **[2]**

28 A, B and C are points on a circle, center O.
The lines TA and TC are tangents to the circle at A and C respectively.
Angle $ATC = 48°$.



(a) Calculate angle AOC.

..... ° **[2]**

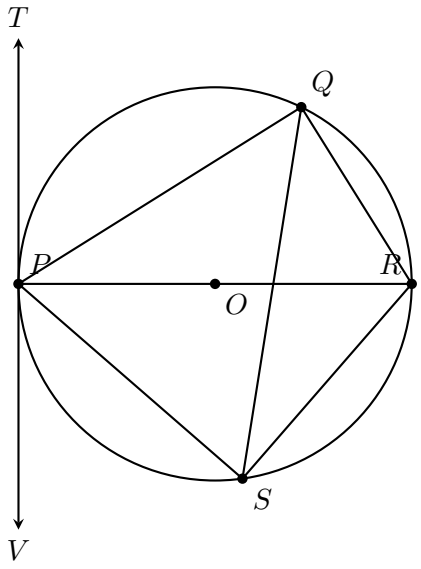
(b) Calculate angle ABC.

..... ° **[1]**

(c) Calculate angle OAC.

..... ° **[2]**

29 P, Q, R and S are points on a circle, center O.
PR is a diameter of the circle.
TPV is a tangent to the circle at P.
Angle $QPR = 32°$ and angle $RPS = 41°$.



(a) Calculate angle PRQ.

Angle $PRQ =$ ..... ° **[2]**

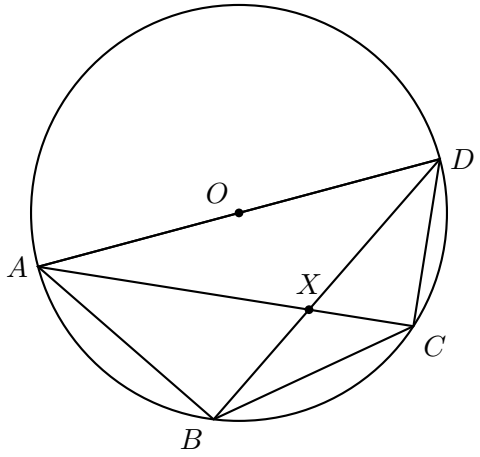
(b) Calculate angle TPQ.

Angle $TPQ =$ ..... ° **[2]**

(c) Calculate angle PQS.

Angle $PQS =$ ..... ° **[2]**

30 AD is a diameter of the circle, centre O.
The points A, B, C, and D lie on the circumference.
The chords AC and BD intersect at X.
Angle $BAC = 32°$ and angle $CAD = 24°$.



(a) Find angle ADC.

..... ° **[2]**

(b) Find angle ABC.

..... ° **[1]**

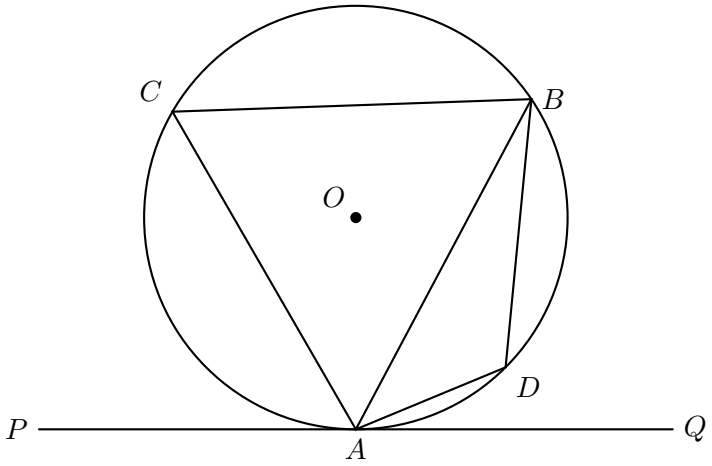
(c) Find angle DBC.

..... ° **[1]**

(d) Find angle AXB.

..... ° **[2]**

31 The diagram shows a circle with centre O.
The line PAQ is a tangent to the circle at the point A.
The points B, C and D lie on the circle.
Angle $QAB = 62°$.



(a) Calculate angle ACB and give a geometrical reason for your answer.

$Angle ACB =$ ..... ° , Reason: ..... **[2]**

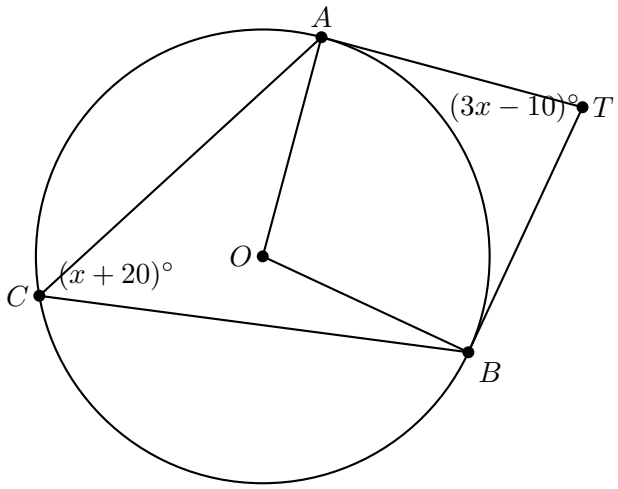
(b) Calculate angle ADB.

..... ° **[2]**

(c) Calculate angle OBA.

..... ° **[2]**

32 The diagram shows a circle, centre O.
A, B and C are points on the circumference.
The lines TA and TB are tangents to the circle at A and B respectively.
Angle $ATB = (3x - 10)^\circ$ and angle $ACB = (x + 20)^\circ$.



(a) Find the value of x.

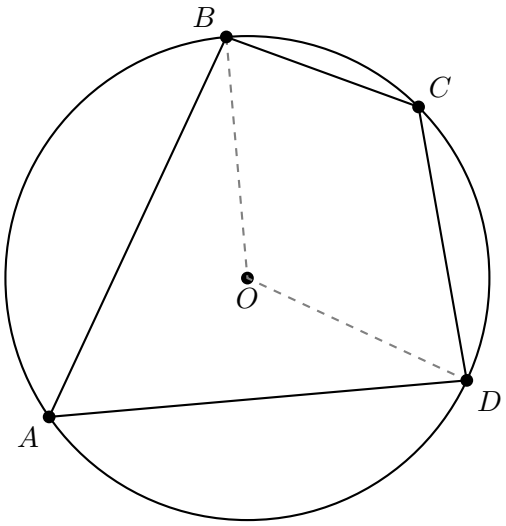
$x =$ ..... **[3]**

(b) Calculate the size of angle OAB.

Angle $OAB =$ ..... ° **[2]**

33 The points A, B, C, and D form a cyclic quadrilateral on a circle with centre O.
The interior angles of the quadrilateral are given in terms of x and y:

$$\angle DAB = (3x + 2y)^\circ$$
$$\angle ABC = (5x + 3y)^\circ$$
$$\angle BCD = (9x + 2y)^\circ$$
$$\angle CDA = (4x + 3y)^\circ$$



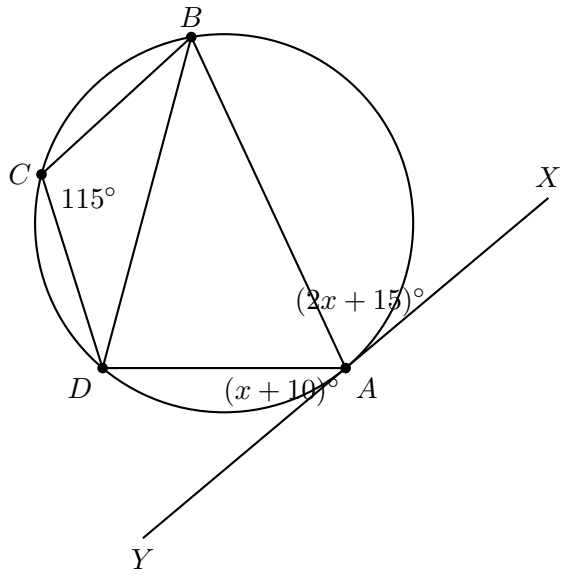
(a) By setting up and solving simultaneous equations, find the value of x and the value of y.

$x =$ ..... , $y =$ ..... **[4]**

(b) Calculate the obtuse angle BOD.

Angle $BOD =$ ..... $^\circ$ **[2]**

34 The points A, B, C and D lie on a circle.
XY is a tangent to the circle at A.
Angle $BCD = 115°$, angle $XAB = (2x + 15)°$ and angle $YAD = (x + 10)°$.



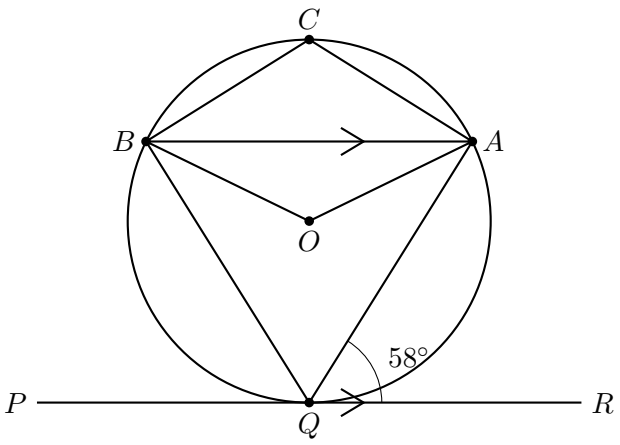
(a) Find the value of x.

$x =$ ..... **[4]**

(b) Calculate angle ABD.

Angle $ABD =$ ..... ° **[2]**

35 A, B, C and Q are points on a circle, center O.
The line PQR is a tangent to the circle at Q.
The chord AB is parallel to PR.
Angle $AQR = 58°$.



(a) **(i)** Find the size of angle ABQ.

..... ° **[1]**

(ii) Give a geometric reason for your answer.

..... **[1]**

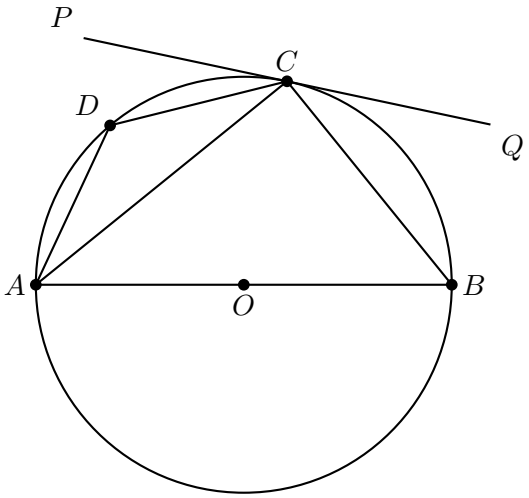
(b) Find angle AOB.

..... ° **[2]**

(c) Find angle ACB.

..... ° **[2]**

36 A, B, C and D are points on a circle, centre O.
AB is a diameter of the circle.
PQ is a tangent to the circle at C.
Angle $CAB = (2x - 5)^\circ$ and Angle $BCQ = (x + 17)^\circ$.



NOT TO SCALE

(a) Set up an equation in terms of x and solve it to show that $x = 22$.
Give a geometrical reason to justify your equation.

[3]

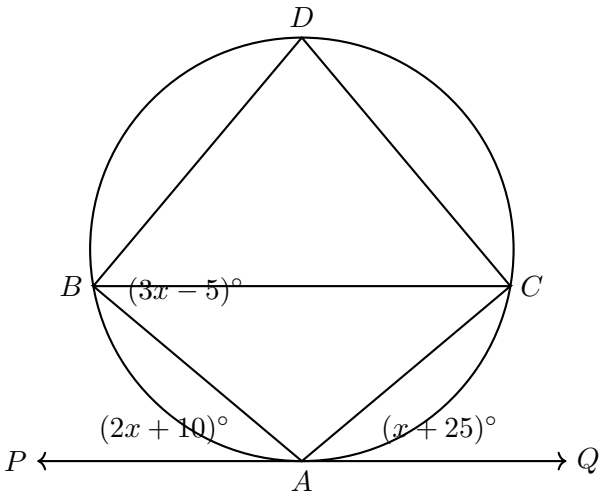
(b) Work out the size of angle CBA.

Angle $CBA =$ ..... ° **[2]**

(c) Find the size of angle ADC.

Angle $ADC =$ ..... ° **[2]**

37 Points A, B, C and D lie on a circle.
The straight line PAQ is a tangent to the circle at A.
Angle $PAB = (2x + 10)°$, angle $ABC = (3x - 5)°$ and angle $CAQ = (x + 25)°$.
D is a point on the major arc BC.



(a) Find the value of x.
Show all your working.

$x =$ ..... **[3]**

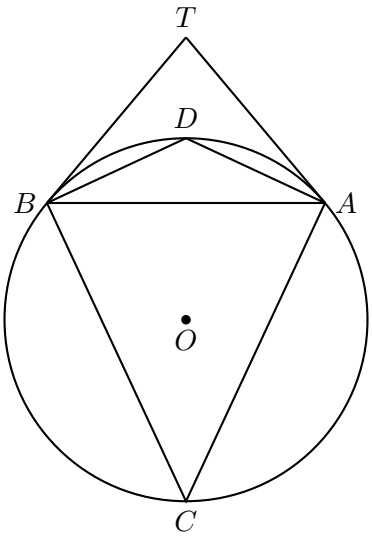
(b) Find the size of angle BAC.

..... ° **[2]**

(c) Find the size of angle BDC.

..... ° **[2]**

38 TA and TB are tangents to a circle, centre O, at the points A and B.
C is a point on the major arc AB and D is a point on the minor arc AB.
Angle $ATB = (x + 10)^\circ$
Angle $TAB = (2y - 10)^\circ$
Angle $ACB = (3y - 40)^\circ$



(a) Find the value of y.

$y =$ ..... **[3]**

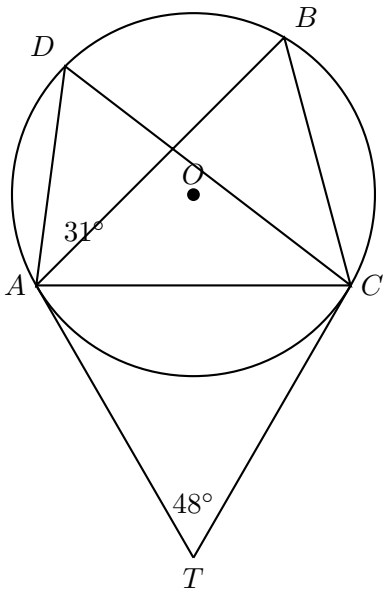
(b) Find the value of x.

$x =$ ..... **[2]**

(c) Find angle ADB.

Angle $ADB =$ ..... $^\circ$ **[2]**

39 A, B, C and D are points on a circle, centre O.
TA and TC are tangents to the circle at A and C.
Angle $ATC = 48°$ and angle $CAD = 31°$.



(a) **(i)** Find angle ABC.

Angle $ABC =$ ..... ° **[2]**

(ii) Find angle ADC.

Angle $ADC =$ ..... ° **[2]**

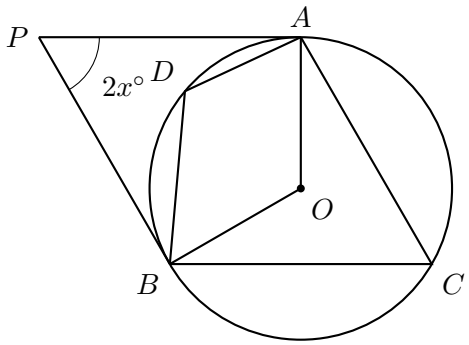
(iii) Find angle ACD.

Angle $ACD =$ ..... ° **[1]**

(iv) Calculate angle AOC.

Angle $AOC =$ ..... ° **[2]**

40 The points A, B, C, and D lie on a circle, centre O.
PA and PB are tangents to the circle at A and B.
Angle $APB = 2x°$.
C is a point on the major arc AB and D is a point on the minor arc AB.



(a) Find an expression, in terms of x, for each of the following angles.
Give each answer in its simplest form.

Angle AOB

..... ° **[1]**

(b) Angle ACB

..... ° **[1]**

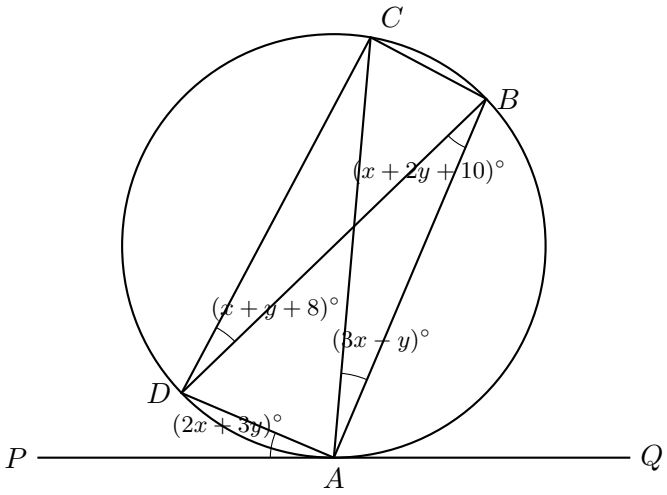
(c) Angle ADB

..... ° **[2]**

(d) Given that angle $ADB = 3 \times$ angle ACB, find the value of x.

$x =$ ..... **[3]**

41 The diagram shows a circle containing the points A, B, C and D.
The straight line PAQ is a tangent to the circle at A.
The chords AC and BD intersect.
Angle $PAD = (2x + 3y)^\circ$ and angle $ABD = (x + 2y + 10)^\circ$.
Angle $BAC = (3x - y)^\circ$ and angle $BDC = (x + y + 8)^\circ$.



NOT TO SCALE

(a) **(i)** Use the circle theorems to write down two equations in x and y.

[2]

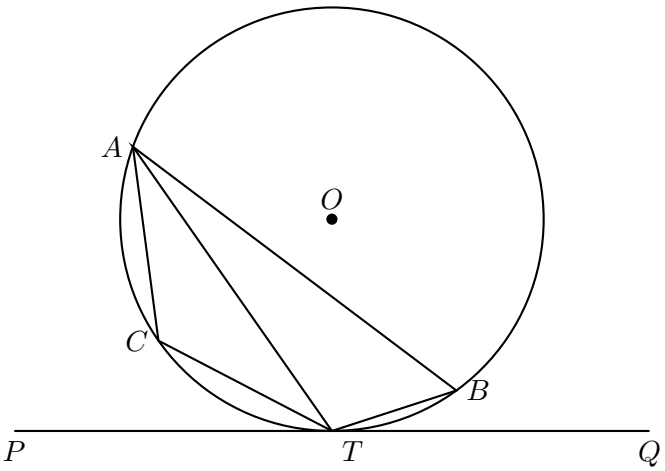
(ii) Solve your equations to find the value of x and the value of y.
Show all your working.

$x =$ ..... , $y =$ ..... **[3]**

(b) Given that angle $ABC = 95^\circ$, work out the size of angle QAB.

Angle $QAB =$ ..... ° **[3]**

42 A, B and C are points on a circle, center O.
The straight line PTQ is a tangent to the circle at the point T.
$\angle ATP = (4x - 5)^\circ$ and $\angle ABT = (2x + 25)^\circ$
$\angle BTQ = (3y)^\circ$ and $\angle BAT = (y + 12)^\circ$



(a) Use the alternate segment theorem to find the values of x and y.

$x =$ ..... , $y =$ ..... **[4]**

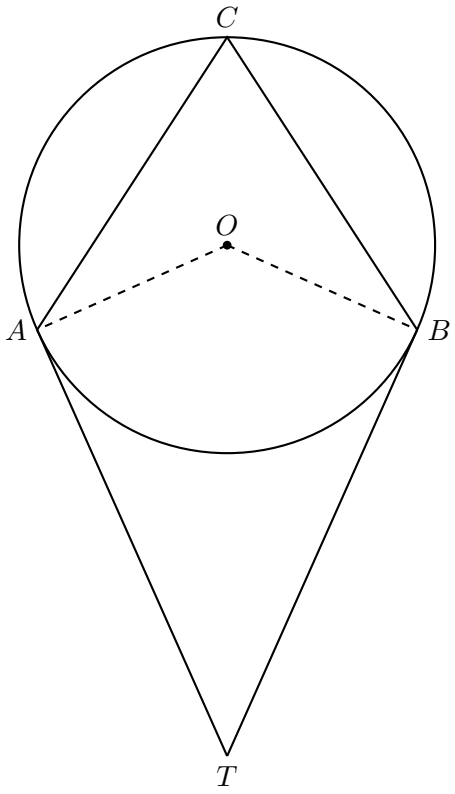
(b) Calculate angle ATB.

..... ° **[2]**

(c) C is a point on the minor arc AT.
Calculate angle ACT.

..... ° **[2]**

43 The diagram shows a circle with centre O.
The lines TA and TB are tangents to the circle at the points A and B.
C is a point on the major arc AB.
Angle $ATB = x°$ and angle $ACB = (2x - 30)°$.



(a) (i) Find the value of x.

$x =$ ..... **[4]**

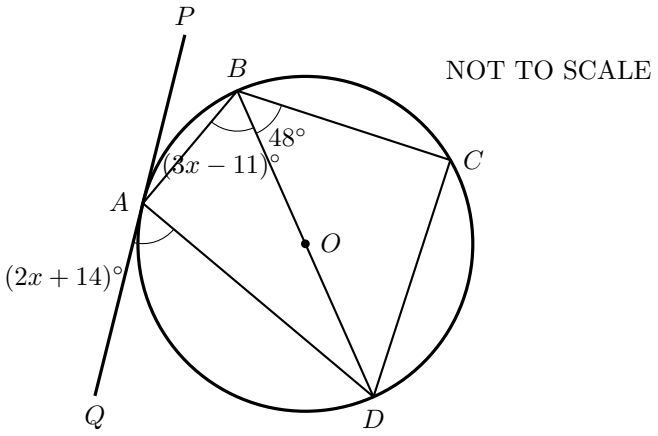
(ii) Calculate angle OAB.

..... ° **[2]**

(b) D is a point on the minor arc AB.
Calculate angle ADB.

..... ° **[2]**

44 The diagram shows a circle, centre O, of radius 9.4 cm. Points A, B, C, and D lie on the circumference, and BD is a diameter of the circle.
The straight line PAQ is a tangent to the circle at A.
Angle $QAD = (2x + 14)°$, angle $ABD = (3x - 11)°$, and angle $CBD = 48°$.



(a) Find the value of x and state the circle theorem you used.

$x =$ ..... , Theorem: ..... **[3]**

(b) Calculate the size of angle ADC, giving a geometrical reason for your answer.

Angle $ADC =$ ..... , Reason: ..... **[2]**

(c) Calculate the length of chord AC, giving your answer correct to 3 significant figures.

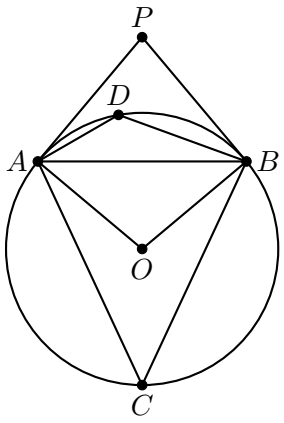
$AC =$ ..... cm **[3]**

(d) Calculate the area of the minor segment bounded by chord AD and the minor arc AD, giving

your answer correct to 3 significant figures.

..... cm^2 **[3]**

45 PA and PB are tangents to a circle, centre O, from a point P.
C is a point on the major arc AB.
Angle $APB = 4x°$.
Angle $ACB = (3x - 10)°$.



(a) Find an expression for angle PAB in terms of x.

..... ° **[3]**

(b) Use the alternate segment theorem to form an equation in x, and show that $x = 20$.

[3]

(c) Calculate angle AOB.

..... ° **[3]**

(d) D is a point on the minor arc AB.
Calculate angle ADB.

..... ° **[3]**